

DEEP LEARNING BASED LIGHT-WEIGHT PROTEIN FUNCTION PREDICTION MODEL

A PROJECT REPORT

Submitted by

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**C.V. RAMAN GLOBAL UNIVERSITY
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BONAFIDE CERTIFICATE

Certified that this project report "**DEEP LEARNING BASED LIGHT-WEIGHT PROTEIN FUNCTION PREDICTION MODEL**" is bonafide work submitted by (Biswa Swarup Tripathy, 7th Semester, Registration No.-2101020745), (Amrutanshu Rath, 7th Semester, Registration No.- 2101020748), (Golap Chand Mallik, 7th Semester, Registration No.- 2101020503) & (Sujit Kumar Sethi, 7th Semester, Registration No.- 2101020692) CGU-Odisha, Bhubaneswar who carried out the project under my supervision.

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CERTIFICATE OF APPROVAL

This is to certify that we have examined the project entitled "**DEEP LEARNING BASED LIGHT-WEIGHT PROTEIN FUNCTION PREDICTION MODEL**" is bonafide work submitted by (**Biswa Swarup Tripathy, 7th Semester, Registration No.-2101020745**), (**Amrutanshu Rath, 7th Semester, Registration No.- 2101020748**), (**Golap Chand Mallik, 7th Semester, Registration No.- 2101020503**) & (**Sujit Kumar Sethi, 7th Semester, Registration No.- 2101020692**), CGU-Odisha, Bhubaneswar.

We hereby accord our approval of it as a major project work carried out and presented in a manner required for its acceptance for the partial fulfillment for **Bachelor Degree of Computer Science and Engineering** for which it has been submitted. This approval does not necessarily endorse or accept every statement made, opinion expressed, or conclusions drawn as recorded in this major project, it only signifies the acceptance of the major project for the purpose it has been submitted.

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DECLARATION

I declare that this project report titled "**Deep Learning Based Light-Weight Protein Function Prediction Model**" submitted in partial fulfillment of the degree of **B. Tech in (Computer Science and Engineering)** is a record of original work carried out by me under the supervision of **Dr. Ashish Ranjan** and has not formed the basis for the award of any other degree or diploma, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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ABSTRACT

Proteins are fundamental biomolecules essential for a wide range of biological processes, yet predicting their functions remains a significant challenge in bioinformatics. This project introduces a novel protein function prediction model, combining dilated convolutional neural networks (Dilated CNNs) with a multi-attention mechanism to efficiently capture both local and global dependencies in protein sequences. The model adopts a lightweight architecture, ensuring computational efficiency while maintaining high predictive accuracy. It leverages protein sequence segmentation, embedding layers, and dilated convolutions with varying dilation rates, combined with attention mechanisms to highlight crucial regions of the sequence for better classification. The model is trained on the Data2017 dataset, which includes protein sequences annotated with Gene Ontology terms, focusing on Biological Process (BP) and Molecular Function (MF) sub-ontologies. Experimental results demonstrate that the proposed model outperforms traditional CNN-based models in terms of precision, recall, and Fmax scores. The use of dilated convolutions allows for the efficient capture of both short- and long-range dependencies, while the multi-attention mechanism enhances the model's ability to focus on critical sequence segments. This approach is designed to be lightweight, making it suitable for deployment on standard and edge computing devices, and provides a scalable solution for protein function prediction in bioinformatics research.

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ABBREVIATIONS

ABBREVIATION	FULL FORM
PFP	Protein Function Prediction
CNN	Convolutional Neural Network
GO	Gene Ontology
BP	Biological Processes
MF	Molecular Functions
ReLU	Rectified Linear Unit
DC-CNN	Dilated Convolution Neural Network
AUPR	Area Under the Precision Recall Curve

Chapter 1

INTRODUCTION

1.1. Protein

Proteins are indispensable biomolecules that form the cornerstone of life, enabling and regulating numerous essential processes in living organisms. They play a pivotal role in biological functions such as metabolic reactions, providing structural integrity, and facilitating immune system responses. Additionally, proteins drive molecular activities, including enzymatic reactions and hormone regulation. Beyond these fundamental roles, proteins are central to biomedical research, serving as crucial components in disease identification and drug development.

Protein research spans multiple modalities, including protein sequences, structural data, and interaction networks. These data forms have enabled significant progress in tasks like protein function prediction, structure modeling, and localization analysis. Of these, protein sequences are the most readily available and extensively studied modality due to their abundance and accessibility. For instance, the UniProt database hosts over 225 million protein sequences, yet only a small subset approximately 1% has been thoroughly annotated and reviewed.

This vast and largely unexplored repository underscores the importance of automated approaches for understanding protein functions. Computational techniques capable of decoding these sequences are essential for advancing our knowledge of protein biology. Such efforts not only deepen our understanding of proteins but also hold the potential to transform applications in biotechnology, medicine, and drug discovery, addressing some of the most pressing challenges in modern biology.

1.2. Problem Specification

The functional behavior of proteins is determined by the complex short- and long-range interactions between amino acids in a sequence. While convolutional neural networks (CNNs) have shown potential in analyzing sequential data, such as protein sequences and natural language. However, their effectiveness is limited to capturing short-range interactions, making them less suitable for modeling the broader dependencies crucial for protein function prediction.

Dilated convolutional networks offer a compelling solution by expanding the receptive field, allowing them to efficiently capture both short- and long-range sequence relationships. These architectures balance computational efficiency with performance, presenting an attractive alternative to the heavily parameterized, multimodal deep learning models currently used for protein function prediction. However, despite their promise, existing approaches often fail to achieve the desired balance between accuracy and resource efficiency, limiting their applicability on standard or low-power devices.

This project proposes a framework that combines dilated convolutional networks with an attention mechanism to enhance protein function prediction. The model adopts a segmentation-based approach, leveraging dilated convolutions to identify both local and global sequence patterns, while the attention mechanism highlights critical regions of the protein sequence, ensuring accurate predictions. Designed to be lightweight, the framework reduces computational complexity while maintaining high performance, making it deployable on standard hardware and edge devices.

The proposed model stands out by significantly reducing trainable parameters compared to existing solutions, achieving superior accuracy with efficient resource usage. By integrating advanced deep learning techniques like dilated convolutions and attention mechanisms, this framework establishes a highly effective and scalable approach to protein function prediction, addressing the growing need for lightweight, sequence-focused solutions in bioinformatics.

Chapter 2

LITERATURE REVIEW

Protein function prediction (PFP) using sequence data has been extensively studied, with numerous models developed to address its inherent challenges. Deepak et al. [1] introduced a λ -scaled attention mechanism to overcome challenges in applying attention-based networks to protein sequences, demonstrating improved performance in protein function prediction with faster convergence and efficient learning. Fahad et al. [2] proposed the MCWS-transformer, incorporating a multi context-window scaled self-attention mechanism to enhance protein sequence modelling, achieving superior predictive performance and outperforming state-of-the-art approaches in protein function prediction tasks. Kumar et al. [3] introduced Lite-SeqCNN, a lightweight, dilated CNN-based architecture for protein function prediction, effectively capturing both short- and long-range interactions with fewer parameters and achieving up to 5% improvement over state-of-the-art models on prominent UniProt datasets. Ranjan et al. [4] proposed the Global-ProtEnc and Global-ProtEnc-Plus methods, leveraging a multi-attention-based multi-aspect network for efficient protein sub-sequence classification, achieving significant improvements in Fmax (+6.50% for BP and +1.90% for CC) on the CAFA3 dataset compared to DeepGOPlus. Hoehndorf et al. [5] introduced DeepGOPlus, a CNN-based protein function prediction model combining sequence similarity and motif detection, achieving competitive Fmax scores on CAFA3 evaluations and outperforming several state-of-the-art methods while offering rapid annotation capabilities.

Zhou et al. [6] proposed DeepGOA, a graph convolutional network (GCN)-based model for protein function prediction, leveraging Gene Ontology (GO) hierarchy and semantic relations. It demonstrated superior performance over state-of-the-art methods in experiments on human and corn datasets. Kumar et al. [7] proposed a CNN-based architecture for protein function prediction by detecting motif information from sub-sequences and annotating full-length sequences. The method outperformed existing approaches like ProtVecGen-Plus, showing notable improvements, particularly for large protein sequences. Sanderson et al. [8] introduced ProteInfer, a deep CNN model for efficient and precise protein function prediction directly from unaligned amino acid sequences, complementing traditional alignment-based methods. Wang et al. [9] proposed a Multi-

label Linear Discriminant Analysis (MLDA) approach for efficient protein function prediction by reducing high-dimensional sequence features and incorporating biological network data for improved classification.

These contributions highlight the growing trend of using deep learning architectures, such as CNNs, transformers, and attention mechanisms, to address challenges in PFP. Our proposed model builds on these advancements by integrating dilated CNNs and a custom attention mechanism to capture intricate relationships in protein sequences efficiently.

Chapter 3

METHODOLOGY

3.1. Overview

This section describes in details the methodology used for protein sequence classification based on their Gene Ontology terms including the details about the dataset and the proposed model architecture.

3.2. Dataset

In this study, the Data2017 dataset, originally curated by Ranjan et al. [27], was utilized for experiments. Derived from the UniProt database, this dataset is specifically designed for protein function prediction tasks, focusing on two sub-ontologies: Biological Process (BP) and Molecular Function (MF). The dataset contains protein sequences annotated with Gene Ontology (GO) terms, which serve as labels for classification.

To ensure robust evaluation and minimize overfitting, the dataset was curated with a stringent separation between training and testing samples. The test set was composed of sequences with less than 50% similarity to those in the training set. Additionally, the dataset incorporates a support threshold of 100, meaning that only GO terms associated with at least 100 sequences were retained. The details about the number of unique sequences in training and testing samples and output classes of dataset is given in Table 3.2.

Table 3.1: Details about the Training and Testing Samples in Dataset

Dataset	Stat	BP	MF
Data2017	Training Samples	43457	32280
	Testing Samples	3359	3132
	Number of Classes	295	135

The Data2017 dataset, containing annotations exclusively for BP and MF tasks, is particularly suited for evaluating models aiming to capture both short- and long-range

dependencies in protein sequences. Its carefully curated structure ensures fair comparisons while reducing bias in protein function prediction.

3.3. Evaluation Metrics

To access the outcomes of experiment, commonly used evaluation metrics were employed like average precision (P_{avg}) (Eq. 1), average recall (R_{avg}) (Eq. 2), F_{max} (Eq. 3). For a given protein sequence i , let L_i represent the actual set of GO terms and E_i represent the predicted set of GO terms. The metrics were calculated as follows:

$$P_{avg} = \frac{1}{n} \sum_{i=0}^n \frac{L_i \cap E_i}{E_i} \quad (\text{Eq. 1})$$

$$R_{avg} = \frac{1}{n} \sum_{i=0}^n \frac{L_i \cap E_i}{L_i} \quad (\text{Eq. 2})$$

$$F_{max} = \max \left\{ \frac{2 * P_{avg} * R_{avg}}{P_{avg} + R_{avg}} \right\} \quad (\text{Eq. 3})$$

3.4. Proposed Lite-SeqCNN + Multi-Attention Architecture

This study introduces a dilated-CNN-based architecture called Lite-SeqCNN, designed to leverage dilated convolutional neural networks (dilated-CNNs) to capture both local and global dependencies within protein sequences. The model processes protein sequences represented as k-mers, extracting key sequence patterns essential for classification tasks. The architecture comprises six primary components: (1) Protein Segmentation, (2) Word Embedding Layer, (3) Dilated CNN Block with Multi-Attention Mechanism, (4) Concatenation Layer, (5) 1-Dimensional Global Average Pooling Layer, and (6) Output Layer. The Figure 3.1 shows the proposed Lite-SeqCNN + multi-attention model architecture. A detailed explanation of these components is provided below.

3.4.1. Protein segmentation

Protein sequences are pre-processed by segmenting them into overlapping k-mers, which serve as input tokens. For this study, $k=4$ was selected to balance vocabulary size and sequence diversity. Smaller values of k (e.g., $k=3$) produce overly limited vocabularies, reducing the model's ability to detect variations, while larger values (e.g., $k=5$) generate overly large vocabularies, making patterns less recurrent. These k-mers are passed to the embedding layer for transformation into numerical representations.

3.4.2. Word embedding layer

This layer transforms protein k-mers into dense, vectorized embeddings that encode meaningful sequence information. The embedding dimension (d) is set to 64, which provides an optimal trade-off between computational complexity and feature representation. To prevent overfitting, a dropout layer with a dropout rate of 0.3 is added after embedding.

3.4.3. Dilated CNN block with multi-attention mechanism

At the heart of the proposed model is the dilated CNN block, which enables the capture of long-range dependencies by introducing gaps (dilations) in the convolutional filter. The architecture comprises five such blocks, each configured with unique dilation rates to extract features across varying contexts. The detail discussion is done in Section 3.5.

3.4.4. Concatenation layer

The outputs from all five dilated CNN blocks are combined into a single feature matrix through concatenation along the feature axis. This layer retains the diverse features extracted by each block, ensuring no information is lost. The resultant feature map has dimensions proportional to $[L \times \text{nb_filters} \times 5]$ where L corresponds to the number of positions in the input sequence after k-mer segmentation and nb_filters is the number of filters used in the CNN block.

3.4.5. 1-dimensional global average pooling layer

Following the concatenation, a global average pooling layer is applied to reduce the feature matrix into a fixed-size vector. This layer replaces traditional fully connected layers, offering improved regularization and computational efficiency. Additionally, a dropout layer with a rate of 0.4 is incorporated to enhance generalization.

3.4.6. Output layer

The final fixed-length vector is passed through a dense layer with a sigmoid activation function, which outputs probabilities for multiple classes in a multi-label classification scenario. The model employs binary cross-entropy as the loss function and utilizes the Adam optimizer with a learning rate of 0.0005 for training.

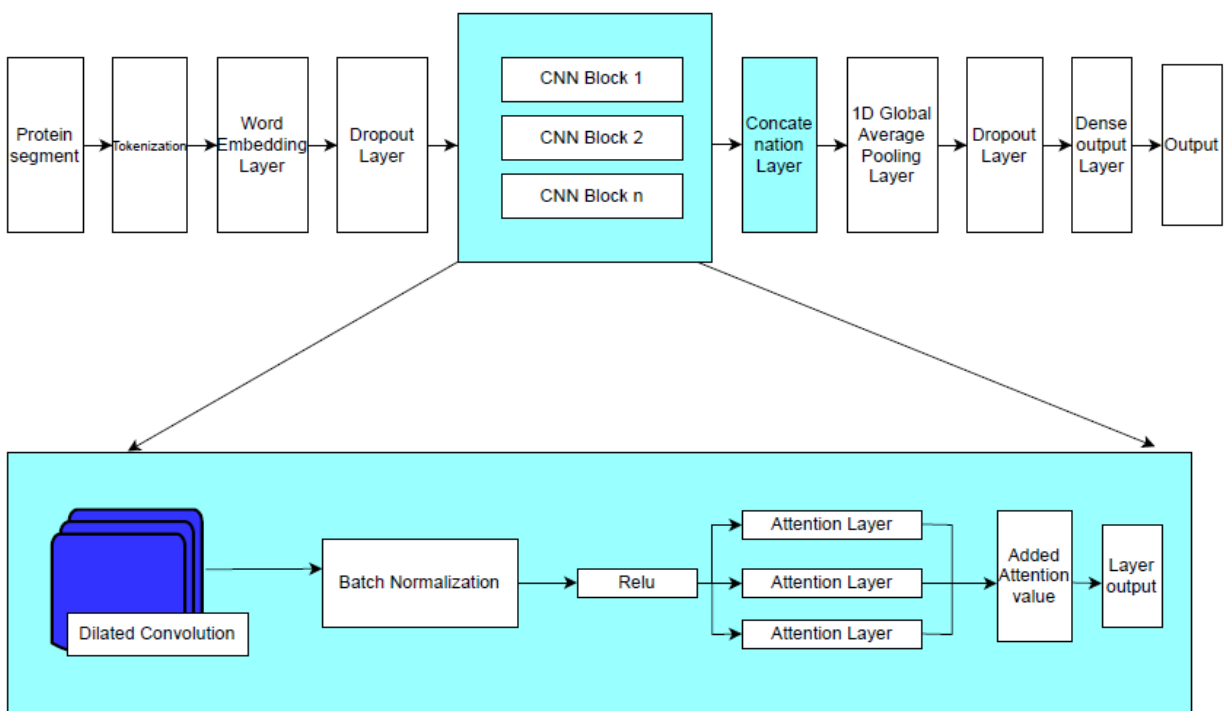


Figure 3.1: Proposed Model with Multi - Attention layers on each CNN block

3.5. Proposed CNN Blocks Architecture

3.5.1. Dilated convolution layer

The Figure 3.1 shows the architecture of each CNN blocks. The proposed CNN block begins with a dilated convolutional layer. Each CNN layer employs a dilation rate, which allows the filter to skip elements in the sequence, significantly expanding the receptive field without increasing the filter size. Such a design enables the model to capture both localized and distant dependencies within the input sequence efficiently. The dilated convolution layer uses multiple number of filters (nb_filter) like 128, 192, 256 to extract diverse features, with a filter size (filter_length) of 6 across all blocks. The dilation rate (dilation) is dynamically varied across blocks (1, 3, 5, 7, and 9) to ensure that interactions at different ranges are captured. The convolution operation is applied with linear activation and padding to preserve the input shape for further operations.

3.5.2. Batch Normalization layer

Following the dilated convolution layer, the output is passed through a batch normalization layer, which normalizes the feature maps to minimize internal covariate shifts during training. This step ensures the stability of feature representations and uses a small epsilon value of $1.1e-5$ to prevent numerical instabilities. Subsequently, a Leaky ReLU activation function introduces non-linearity, enhancing the model's capacity to learn complex patterns. Unlike standard ReLU, the Leaky ReLU alleviates the vanishing gradient problem by allowing a small gradient for negative inputs, with the slope set at $\alpha = 0.2$ which further improves learning efficiency.

3.5.3. Multi-Attention Layer

The block also incorporates a robust multi-attention mechanism, which plays a vital role in enhancing feature representation. This mechanism computes the importance of each step in the feature map using three independent attention heads. Each attention head begins with a dense layer activated by a tanh function, followed by flattening and applying a softmax function to compute attention scores. These scores from the three heads are

combined using an Add layer to create a unified attention vector, which is then repeated to match the number of filters and permuted to align with the feature maps. The attention vector is applied to the feature maps via element-wise multiplication, emphasizing the most relevant features for subsequent layers.

Finally, the feature representation is aggregated using a Lambda layer that performs a summation over the sequence axis. This aggregation step compresses the enhanced feature maps into a concise representation, effectively capturing both local and global features. The architecture utilizes five CNN blocks, each configured with distinct dilation rates (1, 3, 5, 7, and 9) to capture a wide range of interactions. Each block independently processes the input sequence and generates unique feature representations. These outputs are concatenated along the feature axis, creating a rich combined feature vector.

3.6. Prediction of Protein Function

The protein function prediction task in the proposed model is carried out through several stages, starting with data segmentation and followed by feature extraction, model training, and evaluation described in details as follows:

1. **Data Segmentation:** The first step is to segment the protein sequences into smaller chunks of fixed length. This is done using a sliding window approach with an overlap. The segment function divides each sequence into non-overlapping segments of size 200, with a specified overlap 50. This helps capture relevant features from different parts of the sequence.
2. **Feature Extraction (N-grams):** After segmentation, the sequences are transformed into n-grams. N-grams are contiguous sequences of n amino acids, and their occurrence is used to represent the protein sequence. The dictionary function creates a dictionary of all possible n-grams from the training data, which is then used in the nGram function to

convert the protein sequences into numerical representations based on the dictionary. This encoding helps capture the local structure of the sequence.

3. **Model Building (DC-CNN):** The encoded protein sequences are then passed into the DC-CNN (Dilated Convolutional Neural Network) model. The model is composed of multiple convolutional layers with different dilation rates to capture varying levels of context within the sequences. The DC-CNN architecture includes attention mechanisms to focus on the most relevant parts of the sequence. This allows the model to effectively learn from local patterns and global structures simultaneously. The final output of the network is a set of probabilities that represent the likelihood of each protein function class.
4. **Training:** The model is trained on the segmented and encoded sequences, with the output corresponding to protein function labels. During training, the model is optimized using binary cross-entropy loss and evaluated using binary accuracy. Early stopping is employed to avoid overfitting by halting training if validation loss does not improve for a set number of epochs.
5. **Evaluation:** After training, the model's performance is evaluated using test data. The `test_fun` function is used to assess the performance of the model on different test datasets, adjusting the decision threshold to maximize performance. Precision, recall, F1 score, and AUPR (Area Under the Precision-Recall curve) are calculated to evaluate the model's ability to classify protein functions accurately. These metrics provide insights into the model's classification performance.
6. **Final Prediction:** Finally, for new, unseen data, the model predicts the protein function by applying the trained model to the test data. The protein sequences are segmented, n-grams are extracted, and the trained model makes predictions based on the learned patterns.

Chapter 4

RESULTS AND DISCUSSIONS

By varying the different parameters like segment size, overlap size, number of filters used in the evaluation of proposed Lite-SeqCNN + Multi Attention model is done. By using the Eq. 1, Eq.2, Eq.3 the average precision, average recall and Fmax score is determined, respectively. Table 4.1 and 4.2 show the performance comparison between Lite-SeqCNN model and proposed Lite-SeqCNN + multi-attention model, based on the number of filters in each CNN block. By keeping the segment size 200 and overlap 50, three different number of filters 128, 192, 256 are taken for evaluation using both molecular functions (MF) and biological processes (BP) dataset. In the experiment it is observed that, in most of the cases, the CNN blocks having 192 number of filters are performing better than 128 and 256 number of filters. And, the proposed Lite-SeqCNN + multi attention model is having higher precision scores than the Lite-SeqCNN model.

Table 4. 1: Model performance comparison based on number of filters in CNN block

Data2017 MF (Segment Size = 200)				
Model	CNN Filters	Recall	Precision	Fmax
Lite-SeqCNN	128	53.44	78.26	63.51
	192	54.24	78.60	64.18
	256	53.92	77.69	63.56
Lite-SeqCNN + Multi-Attention	128	54.95	78.24	64.56
	192	54.46	80.42	64.94
	256	54.73	78.75	64.48

Table 4. 2: Model performance comparison based on number of filters in CNN block

Data2017 BP (Segment Size = 200)				
Model	CNN Filters	Recall	Precision	Fmax
Lite-SeqCNN	128	42.48	67.81	52.24
	192	43.15	67.78	52.74
	256	42.68	67.90	52.44
Lite-SeqCNN + Multi- Attention	128	43.39	69.26	53.35
	192	42.78	68.10	52.55
	256	43.88	66.42	52.84

The models are performing better classification for molecular function (MF) dataset than the biological processes dataset.

The Table 4.3 shows the performance of proposed model, by keeping the number of filters 192 in each CNN block and the segment size 200, and the effect of overlapping is studied by taking overlapping size 50 and 100.

Table 4.3: Effect of overlapping on model performance

Model	Dataset	Overlap	Recall	Precision	Fmax
Lite-SeqCNN + Multi- Attention	MF	50	54.24	78.12	64.97
		100	57.14	79.62	66.45
	BP	50	42.78	68.10	52.55
		100	45.33	70.57	54.34

It is observed that if we increase the overlapping region the performance also increases. But due to increase in overlapping the number of segments increases due to which the training time increases.

Table 4.4: Performance comparison of proposed model with existing approaches

Datasets		BP		MF	
Sl. No.	Approach	Pavg	Fmax	Pavg	Fmax
1	MLDA [9]	0.56	0.40	0.64	0.50
2	ProtVecGen-120	0.42	0.38	0.56	0.50
3	ProtVecGen-Plus	0.44	0.42	0.62	0.56
4	ProtVecGen-Plus+MLDA	0.69	0.50	0.80	0.61
5	Global-ProtEnc [4]	0.51	0.49	0.57	0.60
6	Global-ProtEnc Plus [4]	0.68	0.54	0.79	0.65
7	Lite-SeqCNN-200 [3]	0.50	0.51	0.60	0.63
8	Lite-SeqCNN-300 [3]	0.49	0.51	0.74	0.64
9	Lite-SeqCNN-400 [3]	0.63	0.52	0.79	0.65
10	Proposed Lite-SeqCNN + Multi-Attention Model - 200	0.68	0.53	0.80	0.65

From the Table 4.4 it is observed that the proposed multi-attention model is having comparatively better performance when compared with the existing approach.

Chapter 5

CONCLUSION

This project presents a lightweight and efficient deep learning model, Lite-SeqCNN + multi-attention, for protein function prediction. By combining k-mer segmentation, dilated convolutional layers, and multi-attention mechanisms, the model effectively captures critical features from protein sequences. The proposed approach demonstrates superior performance compared to existing methods, achieving improved precision and F_max values for both Biological Process (BP) and Molecular Function (MF) tasks. Its lightweight architecture ensures computational efficiency, making it suitable for deployment on standard hardware while maintaining high accuracy. This work contributes significantly to advancing protein sequence analysis and has potential applications in bioinformatics, biotechnology, and drug discovery.

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